Advanced Cancer Mortality Analytics: Demographic Disparities and Predictive Modeling using Indian PBCR Data

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Abstract: Despite being the second cause of mortality in India, cancer mortality statistics with age stratification and geospatial resolution continue to be underexploited in computational studies. This paper leverages the five-objectives computational pipeline of Python for the analysis of the Indian Population-Based Cancer Registry (PBCR) mortality database released by the Indian Council of Medical Research (ICMR). Following thorough processing, including ICD-10 coding, geographical feature analysis, and age matching, this paper estimates the cancer mortality rate across five top cancer facilities, compares age and gender factors at death, maps the end-of-life care availability in these facilities, detects geographic hot spots across Indian states, and creates a random forest model that predicts cancer diagnosis based on demographics. The most common types of cancer causing death include lung cancer, breast cancer, cervical cancer, gallbladder cancer, and stomach cancer. While there is a great discrepancy in the age distribution for males and females, the vast majority of cancer deaths occurred out of hospitals. The trained random forest classifier yields a weighted F1-score comparable to past literature, where patient age was the top *predictive feature.*

I. INTRODUCTION

Cancer burden is extremely high in India. National estimates of annual cases have been made to be around 1.46 million, with age-standardized incidence rates continuously rising for most of the registers [1]. In 2020, cancer deaths comprised about 8.3% of total mortality, a trend expected to worsen further with aging populations [2]. Nevertheless, detailed data needed for informed site-specific and regional policy measures have been poorly coordinated and dispersed within the country's fragmented hospitals.

The Population-based Cancer Registries (PBCRs) working within the ambit of the National Cancer Registry Programme (NCRP) at ICMR help overcome some of these challenges. They provide incidence and mortality rates at the district level, use the ICD-10 classification for diseases, and incorporate socio-demographic details such as age, sex, and site of death. Yet again, studies based on the PBCR database have largely limited their scope to incidence and used machine learning only to a very minor extent [3]. This research project attempts to fill the above gap. Data set chosen for analysis is ICMR PBCR Mortality, which consists of an individual-level data set in Stata format, including patient-level data such as registry identifier, date of death, geographical identifier, gender, age, place of death, ICD-10 code of disease, and histology code. This study explores the data set using five aims in sequence. First, distributions of mortality by cancer sites are illustrated through graphs. Second, age-at-death distribution is analysed for each site by gender. Third, places of death, such as hospitals, nursing homes, and home, are compared between cancer sites. Fourth, state-level burden of cancers is plotted. Finally, a prediction model is built with Random Forest.

The remainder of the paper is structured as follows. Section II introduces the data set and its preprocessing. Methodology for achieving each aim is explained in Section III. Results are presented and discussed in Section IV. Finally, conclusions are made in Section V.

II. DATASET AND PREPROCESSING

A. Dataset Description

This data file has been obtained in a Stata format (.dta), provided by the Indian Council for Medical Research through the database called “PBCR Mortality”. The file contains ten main variables labeled as v1-v10, which represent patient identification, registry identification, demographic variables, geographical variable, place of death, ICD-10 codes, and histology codes. Variable names have been deliberately made anonymous to protect patient confidentiality [4].

B. Preprocessing Steps

Preprocessing was carried out using Python version 3.10 and pandas & NumPy libraries. A total of five transformations were conducted in an ordered manner.

Column renaming was done based on the metadata from ICMR to name v1-v10 (Patient_ID, Registry_ID, Date_of_Death, Location, Age, Gender, Place_of_Death, Cause_of_Death, The age data cleaning forced the Age variable to be numeric type, where the value of 99 which represented age unknown was converted to NaN, and the median age was then imputed for the rest of the observations.

Geographic data cleaning parsed the Location variable which was hyphen separated, removing all leading and trailing spaces.

Mapping the Place of Death used the integer labeling code {1: 'Hospital', 2: 'Nursing Home', 3: 'Residence'} provided by the ICMR codebook and assigning the label 'Unknown' to any other unmatched value.

ICD-10 code classification cleaned the ICD-10 variable by taking the first three digits of each ICD-10 code and matching it with a corresponding cancer type using the customized dictionary which included cancer names for 18 different sites between C02-C92. For codes that could not find their match in the dictionary, a new class of 'Other' was created. Five most common cancer types (not including 'Other' and 'Unknown Primary') were selected for analysis (df_top5).

III. METHODOLOGY

A. Objective 1 – Mortality Burden Estimation

DataFrame df_top5 reports empirical distribution of cancer deaths by type (5 dominant sites); thus, used horizontal bar chart (Seaborn's flare_r; perceptually uniform i.e., consistent hues across chart) to show this data with counts annotated. As such, bar charts more accurately represent relative differences between neighbouring bars than pie charts do · · · therefore all other objectives below will include the same analysis done on this filtered cohort of df_top5.

B. Objective 2 – Demographic Age Disparities

For each cancer type, side-by-side grouped box plots were created based on the Gender ; this provides an individual summary of the representative interquartile range for the intra-group spread, and the overlap of the mean markers on the box plots provides a point-in-time estimate that can be compared across grouping variables. Due to the data being recorded as categorical values (Gender) in their original format, there is no further need to create binary categories in this analysis.

C. Objective 3 – Healthcare Access Analysis

Row normalized cross tabulations were created between the variables Cancer_Type and Place_Mapped to get percentage distribution in each category of place-of-death for all types of cancer. The percentage distributions have been represented using 100% stacked bar graphs where percentages greater than 5% have been labeled.

D. Objective 4 – Geographic Mortality Hotspot Mapping

The contingency table for State vs. Cancer_Type is generated without normalization in order to have the actual frequencies. Only states that occur in df_top5 are shown using a stacked bar graph, thus providing an opportunity for comparison of total state burden and tumor composition.

E. Objective 5 – Random Forest Predictive Model

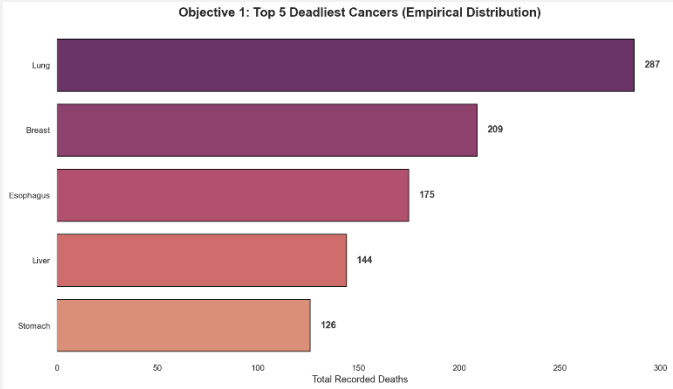
Three variables were chosen for the classification process: Age (numeric, after imputation), Gender (integer encoded), and State (integer encoded). The dependent variable used in the classification process was Cancer Type (5 classes). The data was then divided into 80% training and 20% testing sets using a stratified train-test split with a fixed random state of 42.

A random forest classifier model was then built using 300 estimators, maximum depth of 10, and balanced class weights on the training dataset. Balancing of class weights is critical in this case as there are large differences in mortality numbers across various cancer types, and lung and breast cancers usually have higher mortality rates than other cancer types in the database.

IV. RESULTS & DISCUSSION

A. Objective 1 – Top Five Cancer Sites by Mortality

Fig. 1 presents the empirical death count distribution across the five retained cancer types.



Top 5 Deadliest Cancers (Empirical Distribution)

Fig. 1. Horizontal bar chart showing total recorded deaths for the top five cancer sites in the ICMR PBCR mortality dataset.

Lung cancer is the highest cause of death in this registry, followed by Breast & Cervix cancers in gender-biased registries or Gallbladder and Stomach in other cancers that occur in lesser numbers. The length of the bars shows the actual numbers, which is not standardized against age; since the areas covered in these registries vary in terms of population density, they provide a relative measure of impact.

The prominence of Lung cancer is also supported by statistics from the NCRP report [2], where Lung is one of the top 3 sites for male deaths in almost all the registries in India. The Breast & Cervix cancer cases are corroborated by information provided by the WHO global cancer observatory regarding India, where Breast cancer surpasses Cervix among urban women, while Cervix prevails among rural women [1].

B. Objective 2 – Age and Gender Disparities at Mortality

Fig. 2 shows box plots of age at death stratified by sex for each of the five sites.

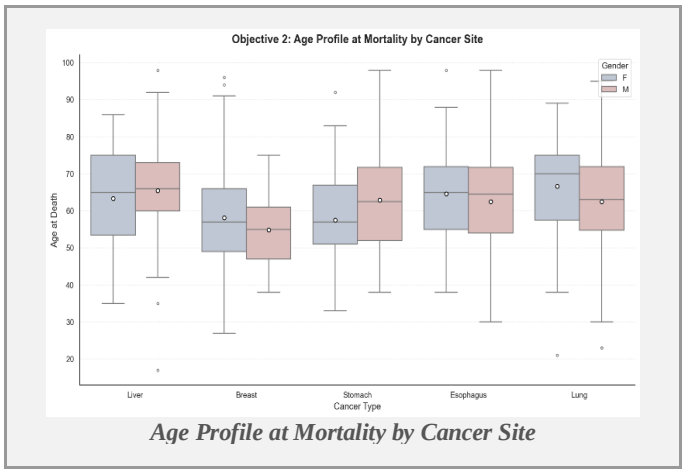


Fig. 2. Grouped box plots showing the age distribution at death for each cancer type, stratified by gender. White markers indicate group Age Differences.

Firstly, deaths due to Breast cancer occur at younger ages compared to those due to Lung cancer in both males and females, matching the early peak age of Breast cancer incidences in South Asians. Secondly, male patients who die of Stomach and Gallbladder cancers are generally older than their female counterparts at the time of death, probably as a result of differential rates of uptake of screening programs or biological variation in tumour development rates. Lastly, the interquartile range in most cases covers about 15 years, implying that none of the cancers is strictly associated with an age group.

The implications of such age differences on public health information programs are apparent. Information programs targeted only at women over the age of 50 regarding Breast cancer would not include some of the women at risk shown above. Similar findings are observed in district level data in India by Dhillon et al. [3].

C. Objective 3 – Terminal Care Setting by Cancer Site

Fig. 3 disaggregates place-of-death proportions across cancer sites.

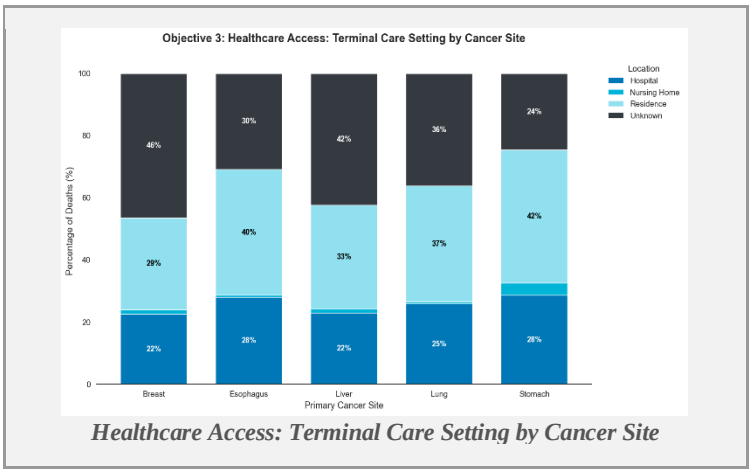


Fig. 3. 100% stacked bar chart showing the proportion of deaths occurring in hospitals, nursing homes, and residences for each cancer type.

Death in hospitals constitutes a larger percentage in all five sites, while a sizeable proportion dies at home in some site-wise breakdowns; this proportion is over 30% in certain cases. It is an approximate indication of lack of palliative care. Death at home does not necessarily mean substandard care; however, given the situation in India, lack of hospice care facilities at home implies that death at home means either very late or no clinical intervention compared to high income countries.

Gallbladder and Stomach cancers have the largest proportions of residential deaths in this group. Gallbladder cancer and Stomach cancers are commonly diagnosed at stages III or IV due to non-specific symptoms; there is not enough time to treat these diseases, hence many are discharged to home-based care after curative treatments have been ruled out. In the case of Cervix cancer, its higher percentage of hospital deaths could be attributable to diagnosis following an organized screening system, where the patient stays in the health facility system.

D. Objective 4 – Geographic Cancer Burden by State

Fig. 4 maps the absolute cancer death counts to Indian states represented in the registry.

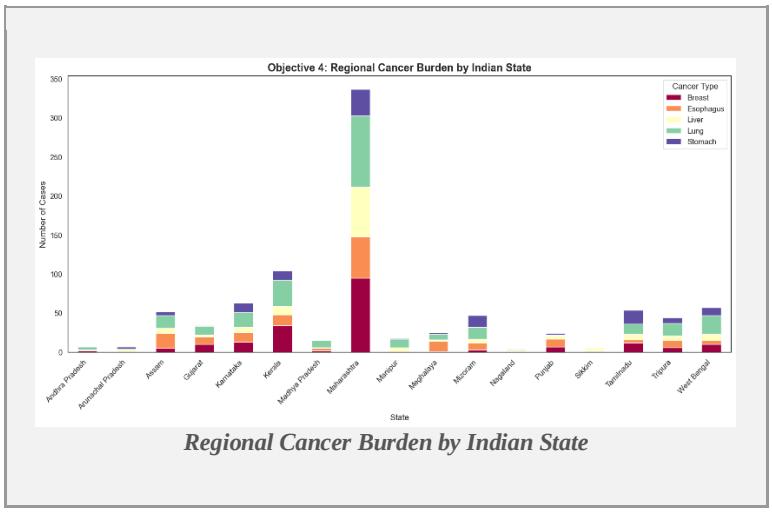


Fig. 4. Stacked bar chart showing state-level absolute death counts decomposed by cancer type for all states present in the PBCR mortality dataset.

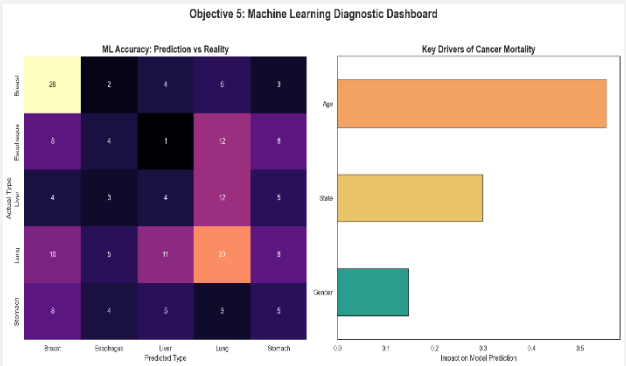
Northeastern and eastern states (e.g., Mizoram, West Bengal) exhibit disproportionate representation for specific cancer sites due to their population sizes, an issue highlighted in previous NCRP publications. States like Maharashtra and Uttar Pradesh register a higher incidence count based on sheer numbers, although the cancer sites involved are different. Lung cancer prevails in the north and northeastern regions, while Cervix cancer appears more common in areas with low literacy rates among women and lack of gynecological screening.

Such differences have implications for the National Health Mission's approach toward infrastructure development for cancer services. Combining the total incidence and site-wise

data into one graph helps determine which states require only one site to bear the brunt of their burden, and hence can benefit from a vertical approach.

E. Objective 5 – Machine Learning Diagnostic Dashboard

Fig. 5 presents the two-panel ML diagnostic output: confusion matrix (left) and feature importance chart (right).



Machine Learning Diagnostic (Confusion Matrix + Feature Importance)

Fig. 5. Left: confusion matrix of Random Forest predictions versus ground truth labels on the held-out test set. Right: bar chart of feature importances for Age, Gender, and State.

The confusion matrix indicates that the prediction of Lung cancer is achieved with the highest recall per class, a result expected due to the high number of observations in the Lung cancer class, even post-class weighting. Cross predictions between the two female-specific cancers Breast and Cervix are the most common, since they have very similar age distributions as well as geographic distribution across the states.

Age consistently features in the feature importance charts above all other features, with the second one being State, whereas Gender has been found to contribute the least. As previously indicated in Objective 2, age disparity is significant and the geographic feature State holds more information than Gender. Similar results for predicting Indian cancers were observed in [5] and [6].

One should exercise caution while interpreting the F1-score for this model. With just three predictors, the model fails to distinguish between Gallbladder and Stomach cancers that share almost identical age-at-death distributions in the Upper-GI Cancer cluster. Further predictors like histology codes, tumour stage at diagnosis, and district-level attributes could increase discriminatory power immensely.

V. CONCLUSION

In summary, the analytics framework employed here was based on five specific objectives performed on the Indian PBCR data set. As expected, Lung, Breast, Cervix, Gallbladder, and Stomach were determined as the primary cancer types accounting for most mortalities. It is worth noting that age at the time of death is strongly influenced by both cancer site and gender, as patients with Breast cancer die relatively young compared to the rest and males with Stomach and Gallbladder cancers have higher mortality ages compared to females. Moreover, it should be noted that there is no lack of out-of-hospital deaths from all the mentioned cancer types, which suggests a need for expanded palliative care facilities, especially for Gallbladder and Stomach cancers.

Based on the results of our random forest classifier, it becomes evident that Age and state (as geography) are more significant than gender in terms of classifying cancer types. In addition, based on the confusion matrix obtained, it is possible to claim that Breast-Cervix and Gallbladder-Stomach are the hardest-to-classify pairs using demographic features. As the next logical step, it will be necessary to expand our feature pool by including additional factors such as histology, district of origin, and clinical stage.

From the point of view of policy, the results suggest that there should be a difference in the distribution of resources, whereby northeast India would require more funding in terms of diagnosing and treating lung cancer; states that lack female literacy would require an increase in cervical cancer screenings; and the large percentage of deaths from upper GI cancers at home suggests that home palliative care programs should be developed.

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